

FENIL PATEL

Chicago, IL

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PROFESSIONAL SUMMARY

Innovative Solutions Architect and Software Engineer with a Master's in Computer Science. Specialized in building scalable Microservices, AI Agents, and Hybrid Database architectures. Expertise in bridging the gap between Data Engineering and Generative AI using LangChain and Kafka, with a proven track record of optimizing high-performance backend systems.

TECHNICAL SKILLS

Languages & Backend: Python (FastAPI), Node.js, JavaScript (ES6+), SQL (MySQL/PostgreSQL), Java.

AI & GenAI: LangChain, OpenAI API, RAG, Vector DBs (Pinecone), E2B Sandbox, Prompt Engineering.

System Architecture: Microservices, Apache Kafka, Redis (Caching), Docker, RESTful APIs, Hybrid DBs.

Frontend & Tools: React.js, Tailwind CSS, Bootstrap, Git/GitHub, AWS (EC2, S3), Linux.

FEATURED ENGINEERING PROJECTS

AutoAnalyst AI (Autonomous BI Agent) | *Python, FastAPI, React, LangChain, E2B* Jan 2026 – Present

- Architected a full-stack AI agent capable of autonomous data analysis, utilizing **LangChain** to interpret complex user queries and execute Python code in a secure **E2B Sandbox**.
- Engineered a secure **FastAPI** backend with OAuth2 authentication and **MongoDB** for per-user data isolation and history persistence.
- Built a programmatic reporting engine that generates "McKinsey-style" executive summaries and renders 3 distinct visualization types (Line, Bar, Donut) into downloadable PDFs.
- Solved LLM hallucination risks by implementing a code-execution environment, ensuring 100% mathematical accuracy in generated KPIs.

Real-Time Market Data Microservice | *FastAPI, Apache Kafka, Redis, Docker* Oct 2025 – Dec 2025

- Developed a high-throughput microservice to process real-time financial data from Alpha Vantage and YFinance APIs.
- Implemented **Redis** for caching frequently requested price data, reducing API latency and external call costs.
- Integrated **Apache Kafka** producer-consumer architecture to stream price events for downstream Moving Average calculations.
- Containerized the entire stack (API, DB, Cache, Zookeeper) using **Docker Compose** for consistent deployment logic.

SmartHomes E-Commerce Platform | *React, Node.js, MySQL, MongoDB* Aug 2025 – Sep 2025

- Designed a scalable full-stack retailer application featuring Role-Based Access Control (RBAC) for Customers, Salesmen, and Store Managers.
- Architected a **Hybrid Database** system utilizing **MySQL** for transactional data (Orders, Inventory) and **MongoDB** for unstructured product reviews.
- Implemented a "Trending Products" dashboard using data visualization charts to track sales performance across different zip codes.
- Built an inventory management system that automatically updates stock levels upon order confirmation via a transactional REST API.

PROFESSIONAL EXPERIENCE

Acodesoft Technologies Gujarat, India

Software Engineer

Jan 2023 – Dec 2023

- Engineered a high-performance data processing engine using **Python**, achieving a **15% reduction** in server-side processing time through algorithm optimization.
- Designed and deployed scalable RESTful APIs to interface with SQL databases, optimizing stored procedures to improve data retrieval speeds by **15%**.
- Spearheaded unit testing protocols, increasing code coverage by **40%** and reducing deployment bugs.

Dynamic Dreamz

Gujarat, India

Data Engineer & Analyst

Jan 2021 – Dec 2022

- Architected an automated ETL pipeline using **Python and SQL** to migrate unstructured data into a centralized Data Warehouse, reducing manual data processing time by **25%**.
- Optimized complex SQL queries for large-scale datasets, resulting in a **20% increase** in query execution speed for real-time reporting.
- Developed interactive dashboards in **Tableau** to visualize KPIs, enabling stakeholders to identify key market trends.

EDUCATION

Illinois Institute of Technology

Chicago, IL

Master of Computer Science (CGPA: 3/4)

May 2025

- **Key Coursework:** Machine Learning, Enterprise Web Applications, Advanced Software Engineering
- **Leadership:** Led a 6-member agile development team to deliver 5 software projects with 100% on-time delivery.